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CSC273-001 Data Structures
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Test Plan

1. Buy 50 shares at $20
    - Adds 50 shares at $20 into our stocks.
    - Print a statement to the user saying how many shares they bought and at what price.
    - Prompt user Menu again.
2. Buy 120 shares at $12
    - Adds 120 shares at $12 into our stocks.
    - Print a statement to the user saying how many shares they bought and at what price.
    - Prompt user Menu again.
3. Buy 20 shares at $35
    - Adds 20 shares at $35 into our stocks.
    - Print a statement to the user saying how many shares they bought and at what price.
    - Prompt user Menu again.
4. Buy 80 shares at $18
    - Adds 80 shares at $18 into our stocks.
    - Print a statement to the user saying how many shares they bought and at what price.
    - Prompt user Menu again.
5. Sell 200 shares at $40
    - All 50 shares from first transaction, gives us a gain of: ($40 - $20) * 50 = $1000
    - All 120 shares from the second transaction, gives us a gain of: ($40 - $12) * 120 = $3360
    - All 20 shares from the third transaction, gives us a gain of: ($40 - $35) * 20 = $100
    - 10 of the 80 shares from the fourth transaction, gives us: ($40 - $18) * 10 = $220
    - Print the current capital gain/Loss from the current transaction. = $4680
    - The total gain/loss variable should be updated to $4680
    - 70 shares left at $18 in our Stocks.
    - Prompt user Menu again.
6. Display total Capital Gain/Loss from all sale transactions
    - Print: Total gain from all sale transactions: $4680
    - Prompt Menu again.
7. Sell 60 shares at $22
    - 60 of the 70 shares left will give us a gain of: ($22 - $18) * 60 = $240
    - Print the current capital gain = $240

- Update the total Capital gain/loss so far to $4920
- 10 shares left at $18 in our Stocks.
- Prompt user Menu again.

8. Display total Capital gain/loss so far
    - Prints: $4920
    - Prompt user Menu again.
9. sell 300 shares at $20
    - print an error message, only 10 of the 300 shares were able to be sold.
    - Will print 0 stocks are left in our deque.
    - Print capital gain made = $20
    - The Total capital gain/Loss should be updated to $4940
    - 0 shares left in our stocks.
    - Prompt menu again
10. Buy 200 shares at $16
    - Adds 200 shares at $16 into our stocks.
    - Print a statement to the user saying how many shares they bought and at what price.
    - Prompt user Menu again.
11. Sell 100 shares at $14
    - 100 shares from the 200 gives us a loss of : (14 - 16) * 100 = $-200
    - Print to the user that they made a loss of $-200
    - The Total capital gain/loss should be updated to $4740
    - 100 shares at $16 left in our stocks.
    - Prompt menu again
12. Display total Realized Capital Gain/Loss so far.
    - Prints: $4740
    - Buy -20 shares at $20
    - Print an error message that shares is invalid.
    - Prompts users with the menu again.
13. Buy 20 shares at $-20
    - Print an error message that the price is invalid.
    - Prompt the users with the menu again.
14. Sell -20 shares at $20
    - Print an error message that shares is invalid
    - Prompt the users with the menu again.
15. Sell 20 shares at $-20
    - Print an error message that price is invalid.
    - Prompt the users with the menu again.
16. Buy -20 shares at $20
    - Print an error message that shares is invalid.
    - Prompt the users with the menu again.
17. Try to enter option 5 at the menu prompt.
    - Print error message to choose one of the options
    - Prompt menu again.

18. Quit
- Program ends.

IMPLEMENTING TEST PLAN IN STOCKS PROGRAM:

BlueJ: Terminal Window - StockCalculator

```
-----------------------------------
Stock Calculator program menu,
OPTIONS:
Buy: type 1
Sell: type 2
Diaply Total Realized Capital Gain/Loss so far: type 3
Quit: type 4
-----------------------------------
1
Option Chosen: Buy
-----------------------------------
How many shares would you like to buy?
50
At what price do you wish to buy them?
20
-----------------------------------
You bought 50 shares at $20.0

-----------------------------------
Stock Calculator program menu,
OPTIONS:
Buy: type 1
Sell: type 2
Diaply Total Realized Capital Gain/Loss so far: type 3
Quit: type 4
-----------------------------------
1
Option Chosen: Buy
-----------------------------------
How many shares would you like to buy?
120
At what price do you wish to buy them?
12
-----------------------------------
You bought 120 shares at $12.0

-----------------------------------
Stock Calculator program menu,
OPTIONS:
Buy: type 1
Sell: type 2
Diaply Total Realized Capital Gain/Loss so far: type 3
```

BlueJ: Terminal Window - StockCalculator

```
Diaply Total Realized Capital Gain/Loss so far: type 3
Quit: type 4
-----------------------------------
1
Option Chosen: Buy
-----------------------------------
How many shares would you like to buy?
20
At what price do you wish to buy them?
35
-----------------------------------
You bought 20 shares at $35.0

-----------------------------------
Stock Calculator program menu,
OPTIONS:
Buy: type 1
Sell: type 2
Diaply Total Realized Capital Gain/Loss so far: type 3
Quit: type 4
-----------------------------------
1
Option Chosen: Buy
-----------------------------------
How many shares would you like to buy?
80
At what price do you wish to buy them?
18
-----------------------------------
You bought 80 shares at $18.0

-----------------------------------
Stock Calculator program menu,
OPTIONS:
Buy: type 1
Sell: type 2
Diaply Total Realized Capital Gain/Loss so far: type 3
Quit: type 4
-----------------------------------
2
Option Chosen: Sell
-----------------------------------
```

Can only enter input while your program is running

BlueJ: Terminal Window - StockCalculator

```
-----------------------------------
How many shares would you like to sell?
200
-----------------------------------
At what price do you wish to sell them?
40
-----------------------------------
You sold 200 shares at $40.0

Your total capitalized gain for this sale: $4680.0
-----------------------------------
Stock Calculator program menu,
OPTIONS:
Buy: type 1
Sell: type 2
Diaply Total Realized Capital Gain/Loss so far: type 3
Quit: type 4
-----------------------------------
3
Option Chosen: Display total capital gain
-----------------------------------
Your total realized capital gain/loss from all sell transactions so far is:
$4680.0
-----------------------------------
Stock Calculator program menu,
OPTIONS:
Buy: type 1
Sell: type 2
Diaply Total Realized Capital Gain/Loss so far: type 3
Quit: type 4
-----------------------------------
2
Option Chosen: Sell
-----------------------------------
How many shares would you like to sell?
60
-----------------------------------
At what price do you wish to sell them?
22
-----------------------------------
You sold 60 shares at $22.0
```

BlueJ: Terminal Window - StockCalculator

```
Your total capitalized gain for this sale: $240.0
-----------------------------------
Stock Calculator program menu,
OPTIONS:
Buy: type 1
Sell: type 2
Diaply Total Realized Capital Gain/Loss so far: type 3
Quit: type 4
-----------------------------------
3
Option Chosen: Display total capital gain
-----------------------------------
Your total realized capital gain/loss from all sell transactions so far is:
$4920.0
-----------------------------------
Stock Calculator program menu,
OPTIONS:
Buy: type 1
Sell: type 2
Diaply Total Realized Capital Gain/Loss so far: type 3
Quit: type 4
-----------------------------------
2
Option Chosen: Sell
-----------------------------------
How many shares would you like to sell?
300
-----------------------------------
At what price do you wish to sell them?
20
-----------------------------------
You were only able to sell part of your stocks
shares sold: 10 out of 300
shares left in stocks: 0
Your total capitalized gain made from this sale: $20.0
-----------------------------------
```




```
-----------------------------------
Stock Calculator program menu,
OPTIONS:
Buy: type 1
Sell: type 2
Diaply Total Realized Capital Gain/Loss so far: type 3
Quit: type 4
-----------------------------------
1
Option Chosen: Buy
-----------------------------------
How many shares would you like to buy?
200
At what price do you wish to buy them?
16
-----------------------------------
You bought 200 shares at $16.0

-----------------------------------
Stock Calculator program menu,
OPTIONS:
Buy: type 1
Sell: type 2
Diaply Total Realized Capital Gain/Loss so far: type 3
Quit: type 4
-----------------------------------
2
Option Chosen: Sell
-----------------------------------
How many shares would you like to sell?
100
-----------------------------------
At what price do you wish to sell them?
14
-----------------------------------
You sold 100 shares at $14.0

Your total capitalized loss for this sale: $-200.0
-----------------------------------
```

BlueJ: Terminal Window - StockCalculator

```
Your total capitalized loss for this sale: $-200.0
-----------------------------------
Stock Calculator program menu,
OPTIONS:
Buy: type 1
Sell: type 2
Diaply Total Realized Capital Gain/Loss so far: type 3
Quit: type 4
-----------------------------------
3
Option Chosen: Display total capital gain
-----------------------------------
Your total realized capital gain/loss from all sell transactions so far is:
$4740.0
-----------------------------------
Stock Calculator program menu,
OPTIONS:
Buy: type 1
Sell: type 2
Diaply Total Realized Capital Gain/Loss so far: type 3
Quit: type 4
-----------------------------------
1
Option Chosen: Buy
-----------------------------------
How many shares would you like to buy?
-20
At what price do you wish to buy them?
20
-----------------------------------
Please re-enter your shares you would like to buy, as 0 or greater, returning to option menu
-----------------------------------
Stock Calculator program menu,
OPTIONS:
Buy: type 1
Sell: type 2
Diaply Total Realized Capital Gain/Loss so far: type 3
Quit: type 4
-----------------------------------
2
Option Chosen: Sell
-----------------------------------
```

Can only enter input while your program is running

BlueJ: Terminal Window - StockCalculator

```
-----------------------------------
How many shares would you like to sell?
-20
-----------------------------------
At what price do you wish to sell them?
20
-----------------------------------
Please re-enter your shares you would like to sell, as 0 or greater, returning to option men
-----------------------------------
Stock Calculator program menu,
OPTIONS:
Buy: type 1
Sell: type 2
Diaply Total Realized Capital Gain/Loss so far: type 3
Quit: type 4
-----------------------------------
2
Option Chosen: Sell
-----------------------------------
How many shares would you like to sell?
20
-----------------------------------
At what price do you wish to sell them?
-20
-----------------------------------
Please re-enter your price you would like to sell your shares at, as 0 or greater, returning
-----------------------------------
Stock Calculator program menu,
OPTIONS:
Buy: type 1
Sell: type 2
Diaply Total Realized Capital Gain/Loss so far: type 3
Quit: type 4
-----------------------------------
1
Option Chosen: Buy
-----------------------------------
How many shares would you like to buy?
20
At what price do you wish to buy them?
-20
-----------------------------------
```

Can only enter input while your program is running

BlueJ: Terminal Window - StockCalculator

```
Quit: type 4
-----------------------------------
2
Option Chosen: Sell
-----------------------------------
How many shares would you like to sell?
20
-----------------------------------
At what price do you wish to sell them?
-20
-----------------------------------
Please re-enter your price you would like to sell your shares at, as 0 or greater, returning to option menu
-----------------------------------
Stock Calculator program menu,
OPTIONS:
Buy: type 1
Sell: type 2
Diaply Total Realized Capital Gain/Loss so far: type 3
Quit: type 4
-----------------------------------
1
Option Chosen: Buy
-----------------------------------
How many shares would you like to buy?
20
At what price do you wish to buy them?
-20
-----------------------------------
Please re-enter your price you would like to buy your shares at, as 0 or greater, returning to option menu
-----------------------------------
Stock Calculator program menu,
OPTIONS:
Buy: type 1
Sell: type 2
Diaply Total Realized Capital Gain/Loss so far: type 3
Quit: type 4
-----------------------------------
4
Option Chosen: Quit
Program Ended
-----------------------------------
```

Can only enter input while your program is running